

Coupled local translation and degradation regulate growth cone collapse

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Evidence abstract

Local translation mediates axonal responses to Semaphorin3A (Sema3A) and other guidance cues.

However, only a subset of the axonal proteome is locally synthesized, whereas most proteins are trafficked from the soma.

The reason why only specific proteins are locally synthesized is unknown.

Here we show that local protein synthesis and degradation are linked events in growth cones.

We find that growth cones exhibit high levels of ubiquitination and that local signalling pathways trigger the ubiquitination and degradation of RhoA, a mediator of Sema3A-induced growth cone collapse.

Inhibition of RhoA degradation is sufficient to remove the protein-synthesis requirement for Sema3A-induced growth cone collapse.

In addition to RhoA, we find that locally translated proteins are the main targets of the ubiquitin-proteasome system in growth cones.

Thus, local protein degradation is a major feature of growth cones and creates a requirement for local translation to replenish proteins needed to maintain growth cone responses.

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Evidence checklist

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2. However, only a subset of the axonal proteome is locally synthesized, whereas most proteins are trafficked from the soma.
3. The reason why only specific proteins are locally synthesized is unknown.
4. Here we show that local protein synthesis and degradation are linked events in growth cones.
5. We find that growth cones exhibit high levels of ubiquitination and that local signalling pathways trigger the ubiquitination and degradation of RhoA, a mediator of Sema3A-induced growth cone collapse.
6. Inhibition of RhoA degradation is sufficient to remove the protein-synthesis requirement for Sema3A-induced growth cone collapse.
7. In addition to RhoA, we find that locally translated proteins are the main targets of the ubiquitin-proteasome system in growth cones.
8. Thus, local protein degradation is a major feature of growth cones and creates a requirement for local translation to replenish proteins needed to maintain growth cone responses.

Source continuity

The canonical evidence above remains the authoritative retrieval target.